

Influenza A+B Lab-On-Time™ Test Device (BMT 51088) CLINICAL VALIDATION REPORT

The purpose of the test is to validate the performance of BMT's Influenza A+B Lab-On-Time™ Test Device and evaluate the usability of this rapid test in the hands of untrained users and their ability to properly perform it based on the provided IFU and to correctly interpret its result.

1. OBJECTIVES

- Direct comparison between the results obtained using the Influenza A+B Rapid Test Device with freshly obtained patient nasal samples and results obtained with matching samples using a lab-based PCR test (the TaqMan™ SARS-CoV-2, Flu A, Flu B Multiplex Assay (AppliedBiosystems, Waltham, MA, USA)).
- Evaluate the usability of BMT's Influenza A+B Lab-On-Time™ Test Device for the purpose of recommending this device for self-testing in the hands of lay untrained users.

2. PERFORMANCE OF TESTING

The test was validated at the Tzafon Medical Center, Poriya, Israel in Feb-Mar 2023 under the supervision of Prof. Avi Peretz. The test was performed according to the test manufacturer's protocol (IFU). The comparison was made with the TaqMan™ SARS-CoV-2, Flu A, Flu B Multiplex Assay (AppliedBiosystems) PCR kit in the lab using the QuantStudio™ 5 Real-Time PCR System (Thermo Scientific, Waltham, MA, USA).

The test was performed by a lab technician, simulating self-testing conditions by collecting a nasal sample from the patient's nose and following the protocol provided in the IFU. A matching nasal specimen was collected and used for PCR validation of the results.

The lab technician didn't have any previous experience with performing this test and was therefore only exposed to and educated by the IFU. This assured that the technician operated the test as a lay person would, thereby simulating self-testing.

3. RESULTS

Table1 – Lab-On-Time Influenza A+B Rapid Test Results compared with the TaqMan™ SARS-CoV-2, Flu A, Flu B Multiplex Assay Test Results

Test		n	Rapid Test Results		% Agreement (With PCR Test Results)
			Negative	Positive	
PCR Test	Negative	100	100	0	100.0%
	Positive	70	17	53	75.7%
Total		170	117	53	

Table2 – Summary of Lab-On-Time Influenza A+B Test Results

Sensitivity	75.7%
Specificity	100.0%
PPV	1.00
NPV	0.86

Next, the Lab-On-Time Influenza A+B Test sensitivity was correlated with the CT values obtained in the Influenza A+B PCR test results for the PCR-positive samples. The results of this comparison are summarized in Table 3.

Table3 – Summary of Lab-On-Time Influenza A+B Test sensitivity according to PCR CT value range

CT Values	Positive	Negative	Total	Sensitivity
<20	10	0	10	100%
20-25	18	0	18	100%
25-30	9	2	11	81.8%
30-35	16	8	24	66.7%
>35	0	7	7	0.0%

A different type of analysis of these results according to CT cutoff values is provided in table 4 below.

Table4 – Summary of Lab-On-Time Influenza A+B Test sensitivity according to CT cutoff values

CT Cutoff value	Test sensitivity
<25	100.0%
<30	94.9%
<35	84.1%

4. OBSERVATIONS AND CONCLUSION

The above evaluation study compared the diagnostic test results obtained using BMT Diagnostics Lab-On-Time Influenza A+B Rapid Test Device with the diagnostic test results obtained with a PCR test (the TaqMan™ SARS-CoV-2, Flu A, Flu B Multiplex Assay (performed in the lab on 170 nasal samples obtained from healthy (100) and influenza patients. From this study, it is concluded that the specificity of BMT's rapid test is 100%, whereas the overall sensitivity was found to be 75.7%.

Further analysis performed according to PCR CT values obtained for nasal samples from influenza patients demonstrated that when considering only CT cutoff values of <35 the overall sensitivity of BMT's rapid test increased to 84.1%.

Furthermore, full correlation was observed between influenza types (A or B) determined by the PCR test and the results obtained with BMT's rapid test.

Finally, BMT's rapid test device has been shown to be very user friendly. Test performance is very intuitive and involves only a single step of sample collection and placement of the test device in the provided, pre-filled reagent tube. Test results were clear and obtained in less than 10 min following nasal sample collection by the device.

It is my professional opinion that BMT Diagnostics Lab-On-Time Influenza A+B Rapid Test Device could be used by the general population as a self-test due to its simplicity, ease of use, and clear results reading.

Signature:



Prof. Avi Peretz

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Appendix – Results

Table 1 – PCR-positive subjects

Sample ID	Antigen Test Result	PCR FLU A	PCR FLU B	CT Value	CT<20	CT 20-25	CT 25-30	CT 30-35	CT>35
FLU104	Positive	Positive	Negative	26.2			P		
FLU105	Positive	Positive	Negative	31.6				P	
FLU106	Positive	Positive	Negative	28.2			P		
FLU107	Positive	Negative	Positive	33.1				P	
FLU108	Positive	Positive	Negative	30.6				P	
FLU109	Negative	Positive	Negative	34.3				N	
FLU110	Positive	Positive	Negative	16.8	P				
FLU111	Negative	Negative	Positive	31.2				N	
FLU112	Negative	Negative	Positive	29.7			N		
FLU113	Negative	Positive	Negative	35.3					N
FLU114	Positive	Positive	Negative	21.8		P			
FLU115	Negative	Negative	Positive	30.2				N	
FLU116	Positive	Positive	Negative	27.4			P		
FLU117	Positive	Positive	Negative	22.8		P			
FLU118	Negative	Positive	Negative	33.9				N	
FLU119	Negative	Negative	Positive	30.8				N	
FLU120	Negative	Negative	Positive	35.2					N
FLU121	Positive	Positive	Negative	24.3		P			
FLU122	Negative	Positive	Negative	36.8					N
FLU123	Positive	Positive	Negative	18.5	P				
FLU124	Positive	Positive	Negative	31.7				P	
FLU125	Positive	Positive	Negative	23.2		P			
FLU126	Positive	Positive	Negative	31.2				P	
FLU127	Positive	Positive	Negative	24.6		P			
FLU128	Positive	Positive	Negative	19.2	P				
FLU129	Negative	Positive	Negative	35.8					N
FLU130	Negative	Positive	Negative	33.9				N	
FLU131	Positive	Positive	Negative	28.5			P		
FLU132	Positive	Positive	Negative	24.8		P			
FLU133	Positive	Positive	Negative	18.6	P				
FLU134	Negative	Positive	Negative	35.8					N
FLU135	Positive	Positive	Negative	24.6		P			
FLU136	Positive	Positive	Negative	31.8				P	

FLU137	Positive	Positive	Negative	30.2				P	
FLU138	Positive	Positive	Negative	21.5		P			
FLU139	Positive	Positive	Negative	30.1				P	
FLU140	Positive	Positive	Negative	24.3		P			
FLU141	Positive	Negative	Positive	19.5	P				
FLU142	Positive	Positive	Negative	22.7		P			
FLU143	Positive	Positive	Negative	24.6		P			
FLU144	Negative	Positive	Negative	36.2					N
FLU145	Positive	Positive	Negative	30.2				P	
FLU146	Positive	Positive	Negative	18.5	P				
FLU147	Positive	Positive	Negative	24.9		P			
FLU148	Positive	Positive	Negative	23.2		P			
FLU149	Positive	Positive	Negative	18.6	P				
FLU150	Positive	Positive	Negative	30.8				P	
FLU151	Positive	Positive	Negative	31.6				P	
FLU152	Positive	Positive	Negative	19.5	P				
FLU153	Positive	Negative	Positive	22.7		P			
FLU154	Positive	Negative	Positive	29.6			P		
FLU155	Negative	Positive	Negative	33.8				N	
FLU156	Positive	Positive	Negative	22.6		P			
FLU157	Negative	Positive	Negative	28.3			N		
FLU158	Positive	Positive	Negative	30.1				P	
FLU159	Positive	Positive	Negative	28.2			P		
FLU160	Positive	Positive	Negative	26.2			P		
FLU161	Positive	Positive	Negative	30.6				P	
FLU162	Positive	Positive	Negative	22.5		P			
FLU163	Positive	Positive	Negative	28.3			P		
FLU164	Positive	Positive	Negative	18.6	P				
FLU165	Positive	Positive	Negative	22.6		P			
FLU166	Positive	Positive	Negative	31.3				P	
FLU167	Negative	Negative	Positive	35.2					N
FLU168	Positive	Positive	Negative	33.6				P	
FLU169	Positive	Positive	Negative	28.6			P		
FLU170	Positive	Positive	Negative	30.8				P	
FLU171	Negative	Positive	Negative	34.8				N	
FLU172	Positive	Positive	Negative	22.3		P			
FLU173	Positive	Negative	Positive	19.5	P				

Table 2 – PCR-negative subjects

Sample ID	PCR Test Result	Antigen Test Result
FLU1	Negative	Negative
FLU2	Negative	Negative
FLU3	Negative	Negative
FLU4	Negative	Negative
FLU5	Negative	Negative
FLU6	Negative	Negative
FLU7	Negative	Negative
FLU8	Negative	Negative
FLU9	Negative	Negative
FLU10	Negative	Negative
FLU11	Negative	Negative
FLU12	Negative	Negative
FLU13	Negative	Negative
FLU14	Negative	Negative
FLU15	Negative	Negative
FLU16	Negative	Negative
FLU17	Negative	Negative
FLU18	Negative	Negative
FLU19	Negative	Negative
FLU20	Negative	Negative
FLU21	Negative	Negative
FLU22	Negative	Negative
FLU23	Negative	Negative
FLU24	Negative	Negative
FLU25	Negative	Negative
FLU26	Negative	Negative
FLU27	Negative	Negative
FLU28	Negative	Negative
FLU29	Negative	Negative
FLU30	Negative	Negative
FLU31	Negative	Negative
FLU32	Negative	Negative
FLU33	Negative	Negative

FLU34	Negative	Negative
FLU35	Negative	Negative
FLU36	Negative	Negative
FLU37	Negative	Negative
FLU38	Negative	Negative
FLU39	Negative	Negative
FLU40	Negative	Negative
FLU41	Negative	Negative
FLU42	Negative	Negative
FLU43	Negative	Negative
FLU44	Negative	Negative
FLU45	Negative	Negative
FLU46	Negative	Negative
FLU47	Negative	Negative
FLU48	Negative	Negative
FLU49	Negative	Negative
FLU50	Negative	Negative
FLU51	Negative	Negative
FLU52	Negative	Negative
FLU53	Negative	Negative
FLU54	Negative	Negative
FLU55	Negative	Negative
FLU56	Negative	Negative
FLU57	Negative	Negative
FLU58	Negative	Negative
FLU59	Negative	Negative
FLU60	Negative	Negative
FLU61	Negative	Negative
FLU62	Negative	Negative
FLU63	Negative	Negative
FLU64	Negative	Negative
FLU65	Negative	Negative
FLU66	Negative	Negative
FLU67	Negative	Negative
FLU68	Negative	Negative
FLU69	Negative	Negative
FLU70	Negative	Negative
FLU71	Negative	Negative
FLU72	Negative	Negative

FLU73	Negative	Negative
FLU74	Negative	Negative
FLU75	Negative	Negative
FLU76	Negative	Negative
FLU77	Negative	Negative
FLU78	Negative	Negative
FLU79	Negative	Negative
FLU80	Negative	Negative
FLU81	Negative	Negative
FLU82	Negative	Negative
FLU83	Negative	Negative
FLU84	Negative	Negative
FLU85	Negative	Negative
FLU86	Negative	Negative
FLU87	Negative	Negative
FLU88	Negative	Negative
FLU89	Negative	Negative
FLU90	Negative	Negative
FLU91	Negative	Negative
FLU92	Negative	Negative
FLU93	Negative	Negative
FLU94	Negative	Negative
FLU95	Negative	Negative
FLU96	Negative	Negative
FLU97	Negative	Negative
FLU98	Negative	Negative
FLU99	Negative	Negative
FLU100	Negative	Negative
FLU101	Negative	Negative
FLU102	Negative	Negative
FLU103	Negative	Negative